Supplementary Figures

Fig. S1 | *tbd* Map of 35 islands surveyed by CREP across the tropical Pacific.

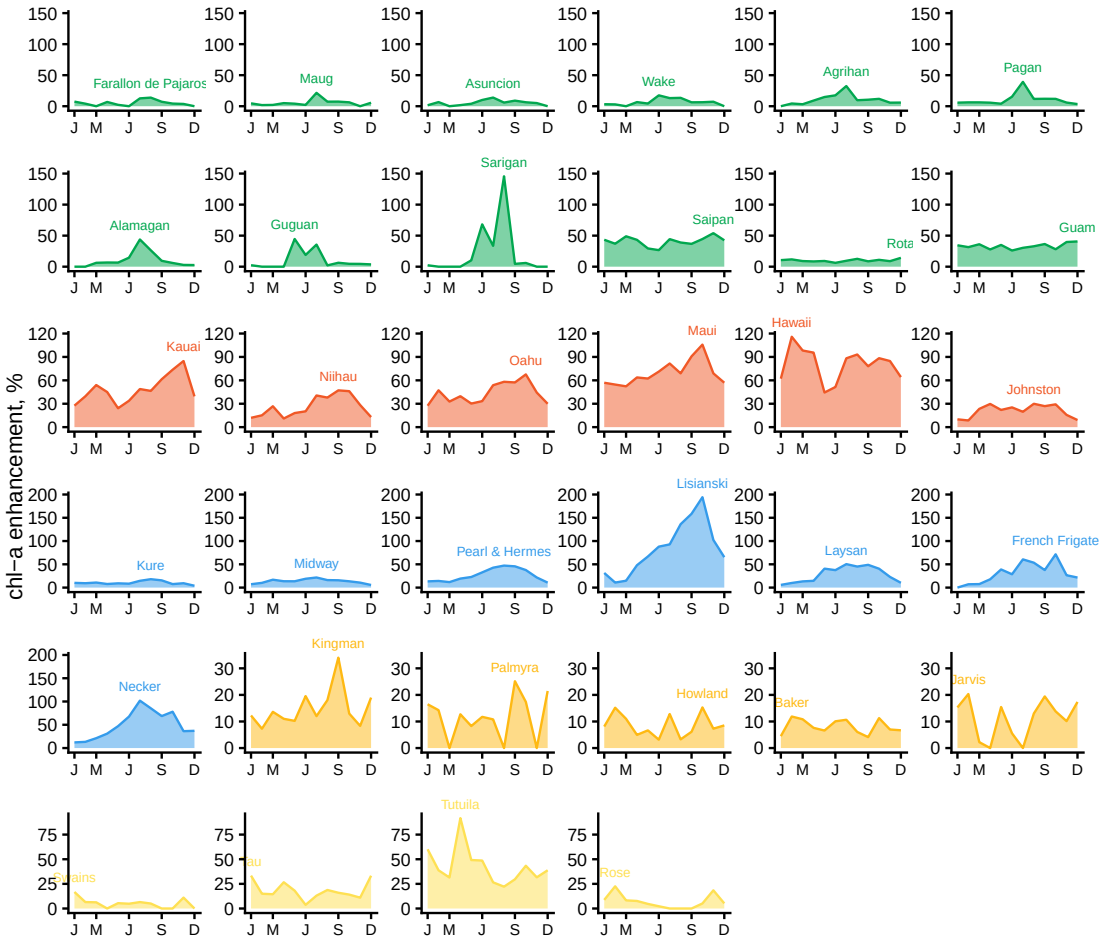

Fig. S2 | Seasonal variation in IME strength at each island complex. Lines are mean-monthly chl-a enhancement from January to December, ordered and coloured by island group.

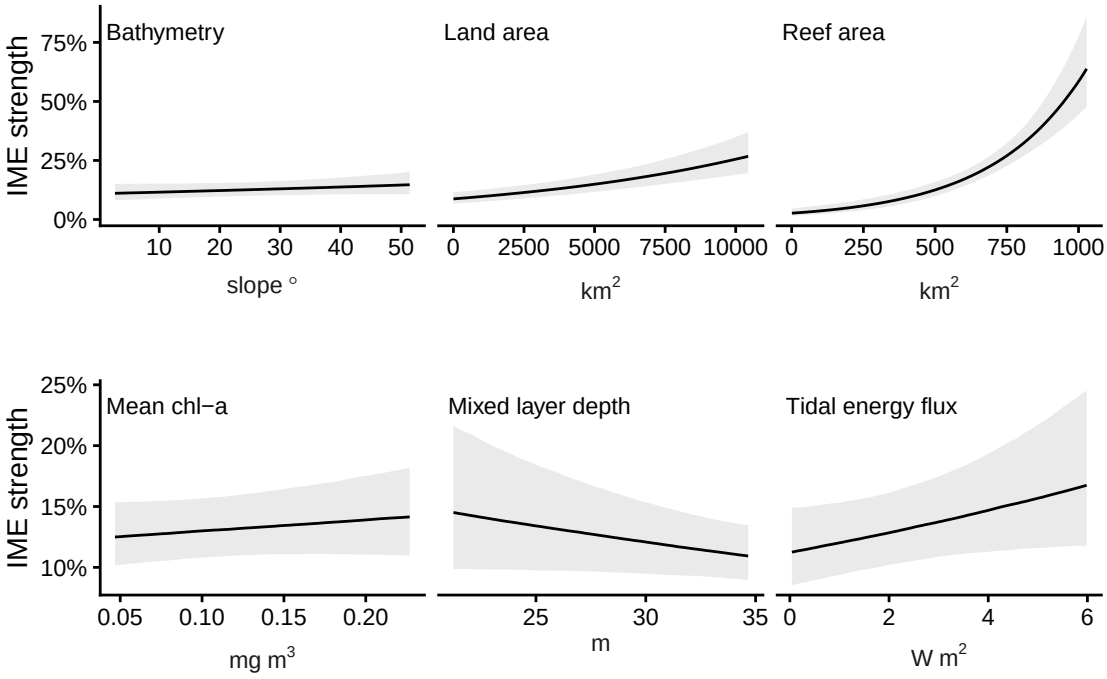

Fig. S3 | Spatial effects on chl-a enhancement for geomorphic features (top) and average oceanographic conditions (bottom) Lines are median posterior predictions (± 95% certainty intervals). Note different y-axis values between static and dynamic covariates.

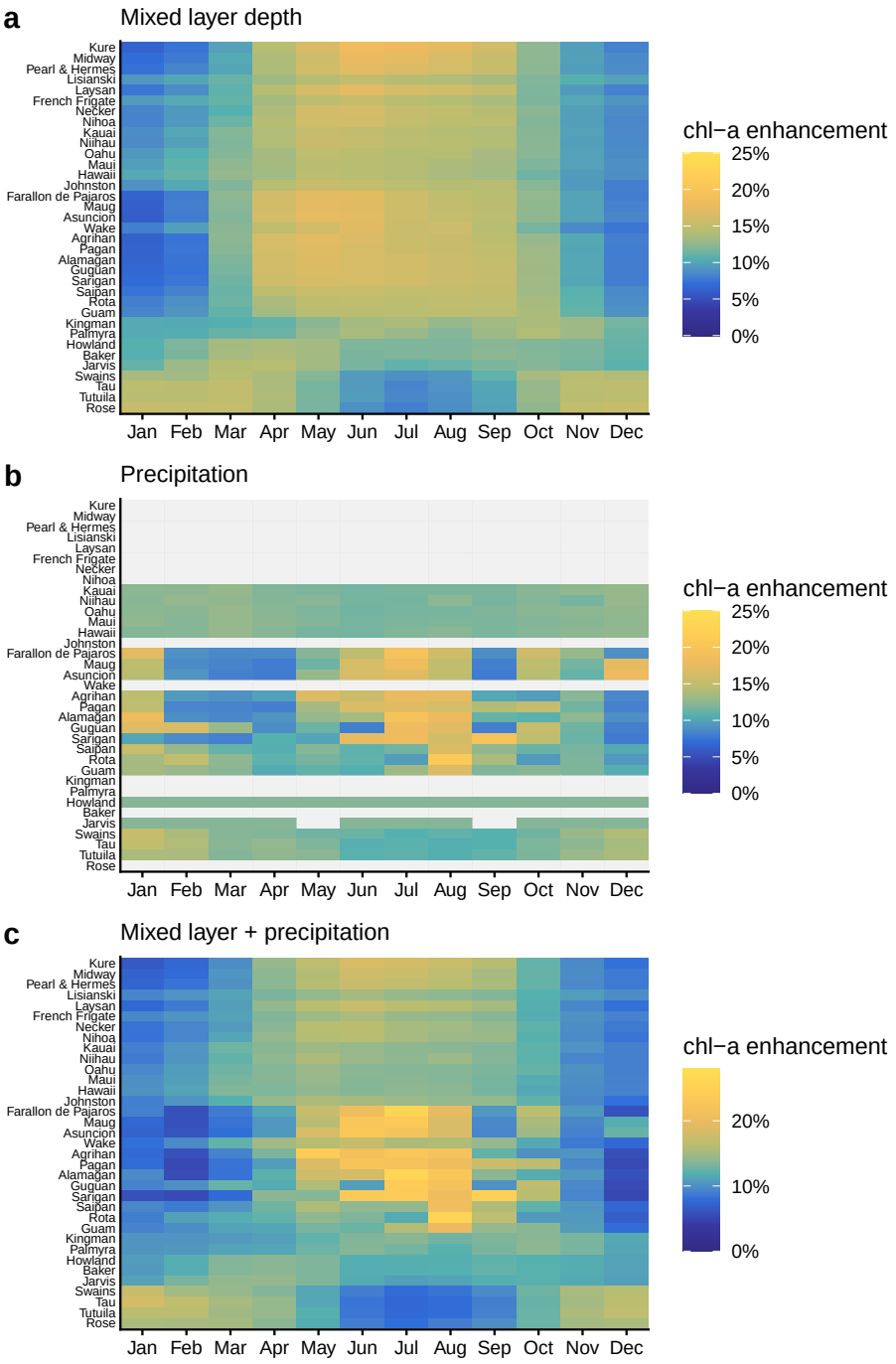

Fig. S4 | Heatmap of seasonal effects on chl-a enhancement for a mixed layer depth, b precipitation and c both covariates combined. Islands with less than 1 mm monthly rainfall are shown in grey. All values are median posterior predictions excluding effects of other covariates.

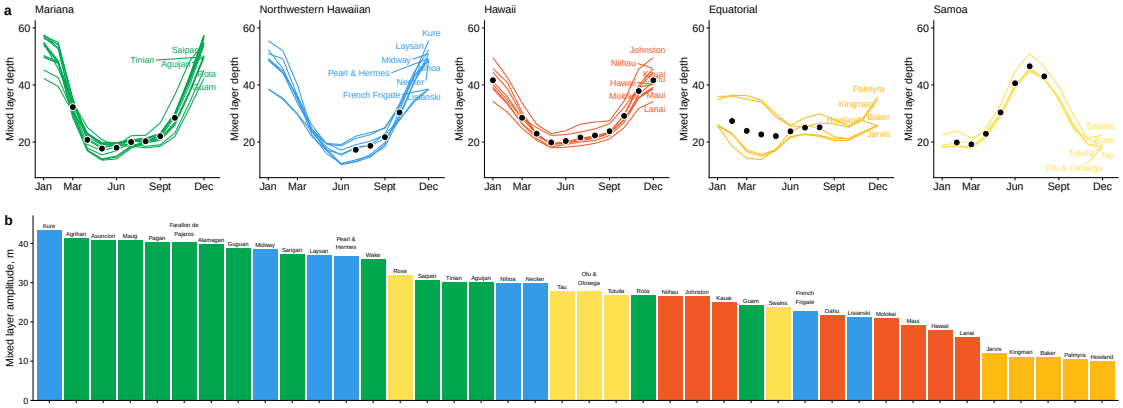

Fig. S5 | Seasonal variability in the mixed layer depth at each island. In a, lines are predicted mixed layer depth at each island from a general additive model of time and month, overlaid with points indicating timing of coral reef fish surveys. Predicted values were used to estimate b the mixed layer depth seasonal amplitude at each island, or the range of depths experienced by each island from Jan to Dec.

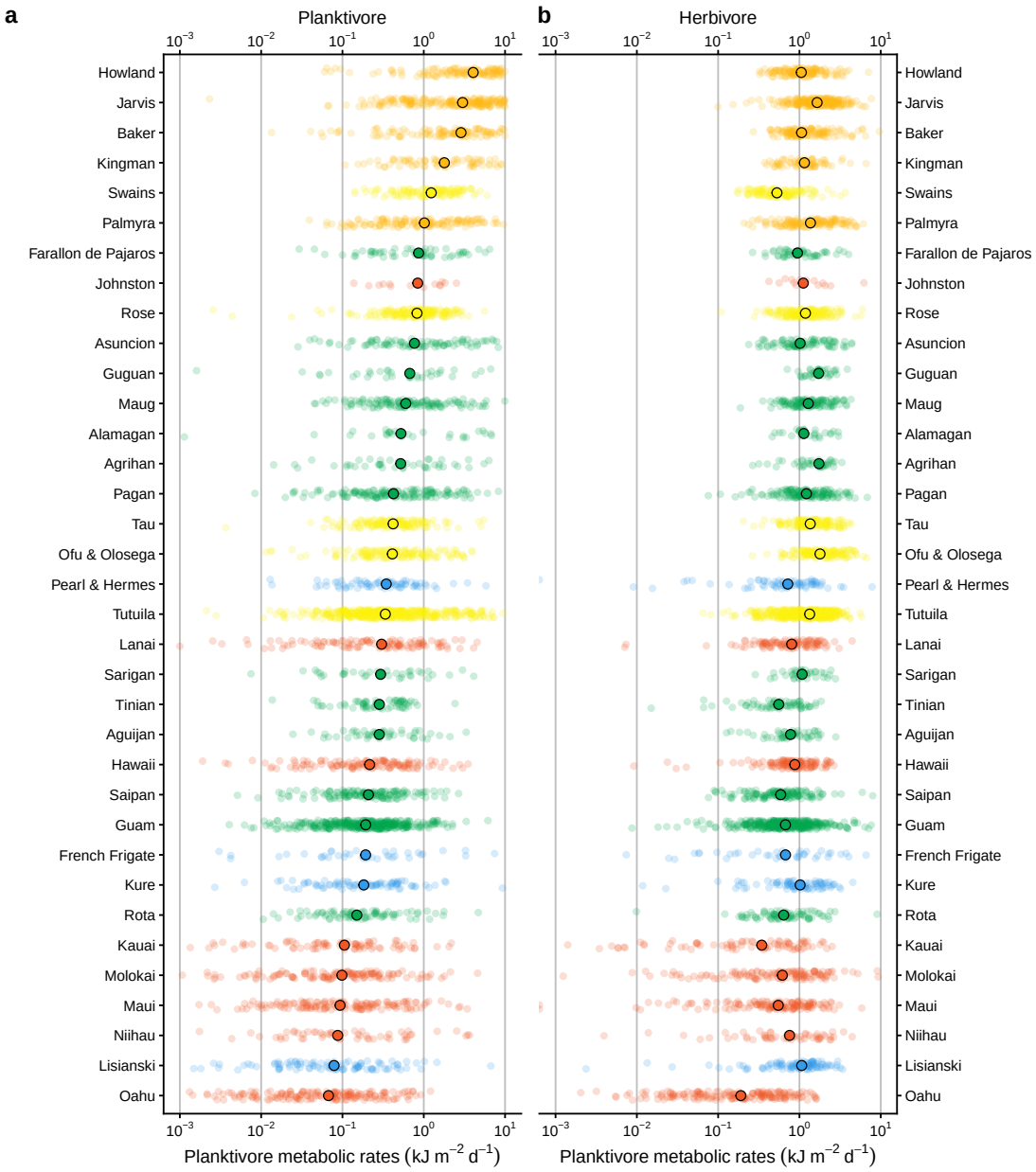

Fig. S6 | Metabolic rate of reef-associated planktivores on coral reefs at 39 Pacific islands and atolls. Each point is a site-year combination (n = XX).